

NEO Mathematics Supporting Adult Guidance

Lesson 3 - Finding a Shorter Side

Connection before curriculum. Always.

This lesson follows directly from the previous hypotenuse lesson. The learner may be expecting to add because that was the successful route last time. The central teaching move is to help them notice that the unknown side has a different role now: it is one of the shorter sides, not the hypotenuse.

Adult role

- Begin with the diagram, not the formula.
- Ask: Where is the right angle?
- Ask: Which side is opposite it?
- Ask: Is the unknown side the hypotenuse or a shorter side?
- Only after that, support the algebraic route.
- Encourage the learner to use the Socratic Prompt before opening the Hint.

Core misconception to expect

A common error is to add the known side squares automatically. This is understandable because the previous lesson used addition to find the hypotenuse. In this lesson, the learner needs to see that the hypotenuse square is already the total, so the known shorter-side square must be subtracted to find the remaining square area.

Useful language

- The hypotenuse square is the whole square-area total.
- The two shorter-side squares combine to make that total.
- If one shorter-side square is missing, subtract the known shorter-side square from the hypotenuse square.
- The square root turns the square area back into a side length.

Interactive support

The Focus this step button does not check an answer. It simply chooses which step the Socratic Prompt and Hint will support. This is intended to reduce cognitive load and let the learner ask for help with one precise decision.

Scratchpad standard

The full NEO Maths Scratchpad is present and should remain available as an optional regulation and reasoning tool. Toolbar order is Pen, Line, Colour, Width, Eraser, Undo, Redo, Sticky Note, Background. Erasing learner marks should not erase the selected background.

When to pause

If the learner is mixing up addition and subtraction, pause the calculation and return to the area meaning: the hypotenuse square is the total. Ask the learner to draw or point to the known part and the missing part before continuing.

Final evidence

A secure final response should identify the hypotenuse, identify the missing shorter side, rearrange the theorem correctly, substitute values, square, subtract and square-root. Mathematical clarity matters more than speed.